

$$1. f: \{0,1\}^2 \rightarrow \{0,1\}^2$$

x	$f(x)$
00	10
01	11
10	10
11	11

$$f(x) = f(y)$$

$$f(00) = f(10)$$

$$\begin{array}{r} 00 \\ \oplus 10 \\ \hline 10 \end{array} \quad \begin{array}{r} 01 \\ \oplus 11 \\ \hline 10 \end{array}$$

$S=10$ and is of quantity 2.

2. Here's the situation:

- Small problem size and a very specific function: If the size of the binary strings (n) is very small (like $n = 2$ or 3), a classical computer might just stumble upon two inputs that satisfy the condition ($f(x) = f(y)$) by pure chance.
- Unreliable and not guaranteed: This lucky guess scenario wouldn't be reliable. There's no guarantee the classical computer will find the right pair quickly, especially as the problem size (n) increases. In contrast, the quantum computer's efficiency is guaranteed to be much better, regardless of the specific function (f).

Here's why quantum computing with Simon's algorithm is generally considered superior:

- Scalability: As the problem size (n) increases, the number of queries needed by a classical computer explodes exponentially. The quantum computer, on the other hand, maintains its logarithmic efficiency (proportional to n), making it significantly faster for practical problem sizes.
- Guaranteed Efficiency: The quantum approach offers a definitive solution path with a small number of queries, regardless of the specific function (f). This makes it reliable and predictable.